

Rings: An Algebraic Structured Peer-to-Peer Network for Decentralized Identity and Resource Ownership

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Abstract

Rings formalizes an implemented structured peer-to-peer network as a carrier-separated authority algebra. The motivating failure mode is authority collapse:

$$\text{root}(id, data, route) \in \{\text{server}, \text{dns}, \text{relay}, \text{platform}\}.$$

Given an overlay, namespaces, protocol effects, and cryptographic sessions, the problem is to construct a total owner map while preventing non-placement carriers from changing ownership:

$$\text{Rings} = (R, \mathcal{R}_N, \text{Set}/R, \Omega_N, \mathbb{T}_A^\mathcal{E}, \Pi, \mathbb{G}, \mathcal{X})$$

with

$$\begin{aligned} R &= \mathbb{Z}/2^{160}\mathbb{Z}, \\ \mathcal{R}_N &= \text{Correct-Chord overlay object}, \\ \Omega_N(X, H) &= (X, \text{owner}_N \circ H), \\ \tau_\nu &: S_\nu \times I_\nu \rightarrow \mathbb{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu). \end{aligned}$$

$$\text{Claim}_{core} = \text{WellDef}(\Omega_N) \wedge \text{CarrierSep} \wedge \text{NonInterfere} \\ \wedge \text{MorphismPreserve} \wedge \text{EffectTyped},$$

$$\text{Claim}_{ledger} = \text{EvidenceScoped}_\Gamma \wedge \text{LedgerDisc}_{Rings}.$$

The contribution is the algebra and its claim ledger:

$$\begin{aligned} \text{Established} &= \text{Core}_{Rings} \wedge \text{LedgerDisc}_{Rings}, \\ \text{Eval}_{reported} &\subset \text{ReportedOnly}, \\ \text{PerfClaim}_R &\in \text{Obligation}. \end{aligned}$$

I. Introduction

Centralized platforms are efficient deployment mechanisms but poor authority carriers: identity, naming, storage, and traffic control collapse to service operators [1]–[3]. Rings studies the following algebraic question:

$$\begin{aligned} \text{Own}(d) &= k_d, \\ \text{Place}_\nu(x) &= \text{owner}_N(H_\nu(x)), \\ \text{Run}(\mathcal{P}_\nu) &= \tau_\nu : S_\nu \times I_\nu \rightarrow \mathbb{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu), \\ \text{RootOfAuthority} &\neq \text{ApplicationServer}, \\ \text{CryptoCarrier} \cap R &= \emptyset. \end{aligned}$$

Chord and CorrectChord supply the routing substrate; DID-style proofs supply self-certifying identity; WebRTC and WebAssembly supply runtime interpretation; CRDTs supply one merge algebra. None alone gives a typed law

for owner, carrier, and evidence scope:

$$\begin{aligned} \text{Chord/CorrectChord} &\Rightarrow \text{ringLookup} \not\Rightarrow \text{resourceAuthority}, \\ \text{DID} &\Rightarrow \text{keyBoundName} \not\Rightarrow \text{overlayOwner}, \\ \text{WebRTC} &\Rightarrow \text{browserTransport} \not\Rightarrow \text{protocolLaw}, \\ \text{CRDT} &\Rightarrow \text{mergeLaw} \not\Rightarrow \text{placementLaw}. \end{aligned}$$

The contribution is a composition theorem plus a ledger discipline. The Chord identifier space is the additive cyclic group for placement and interval order; signatures, threshold schemes, encryption, and zero-knowledge protocols remain in their own fields or groups:

$$\begin{aligned} R &= \mathbb{Z}/2^{160}\mathbb{Z}, \\ \text{CC} &\in \{\mathbb{F}_q, \mathbb{G}, \mathbb{Z}_q\}, \\ R \cap \text{CC} &= \emptyset. \end{aligned}$$

$$\begin{aligned} C_1 &= \Omega_N : \text{Set}/R \rightarrow \text{Set}/N, \\ C_2 &= \text{CarrierSep} \Rightarrow \text{NonInterfere}, \\ C_3 &= \text{Admit}_R(c, k) \Rightarrow \text{Support}_R(c, k), \\ C_4 &= \text{ReportedOnly} \not\Rightarrow \text{PerfClaim}_R. \end{aligned}$$

These four claims define the paper's scientific unit.

II. Problem Statement and Design Goals

Existing decentralized systems solve orthogonal sub-problems. The missing object is the commuting authority layer between identity, placement, execution, and evidence:

$$\begin{aligned} \mathcal{N}_R &= \{(X_\nu, H_\nu)\}_\nu, \\ \mathcal{P}_R &= \{\mathcal{P}_\nu\}_\nu, \\ \text{Input} &= (\Pi, \mathcal{R}_N, \mathcal{N}_R, \mathcal{P}_R, \mathcal{X}), \\ \text{Need} &= \begin{cases} \text{Owner} : X_\nu \rightarrow N, \\ \text{Run} : \mathcal{P}_\nu \rightarrow \text{Effects}, \end{cases} \\ \text{Forbidden} &= \begin{cases} \text{CryptoCarrier} = R, \\ \text{Evidence}_\Gamma = \text{UniversalClaim}. \end{cases} \end{aligned}$$

Obligation 1 (Design target).

$$\begin{aligned} G_{id} &= \text{SelfCertID} \wedge \text{DelegatedSession}, \\ G_{own} &= \text{StructuredOwner}, \\ G_{io} &= \text{BrowserNativeTransport}, \\ G_{eff} &= \text{TypedProtocolEffect}, \\ G_{cr} &= \text{CarrierCorrectCrypto}, \\ G_{rep} &= \text{BoundedRepairEvidence}, \\ G_{bd} &= \text{ExplicitClaimBoundary}, \\ \text{Rings} &\models G_{id} \wedge G_{own} \wedge G_{io} \wedge G_{eff} \\ &\quad \wedge G_{cr} \wedge G_{rep} \wedge G_{bd}. \end{aligned}$$

Definition 1 (Methodology and claim ledger).

$$\begin{aligned} \text{Method} &= \text{Mot} \rightarrow \text{Spec} \rightarrow \text{Algebra} \rightarrow \text{ImplMap} \\ &\quad \rightarrow \text{Evidence}_\Gamma \rightarrow \text{Obligation}, \\ \Gamma &= (|N|, \text{topology}, \text{scheduler}, \text{fault}), \\ \text{ClaimKind} &= \text{Spec} \cup \text{Impl} \cup \text{Evidence}_\Gamma \cup \text{Obligation}, \\ \text{Ev}_\Gamma &= \text{Evidence}_\Gamma, \\ \text{Prop}_R &= \text{dom}(\text{Ledger}_R), \\ \text{Support}_R &: \text{Prop}_R \times \text{ClaimKind} \rightarrow \{\top, \perp\}, \\ \text{Admit}(c) &\Leftrightarrow \exists k \in \text{ClaimKind}. \text{Admit}_R(c, k). \end{aligned}$$

Definition 2 (Global symbols and non-claims).

$$\begin{aligned} \text{NS} &= \{D, V, M, H, U, Q, W\}, \quad \text{VID} = V, \\ \text{Time} &= \mathbb{T}, \quad \text{Ctx} = B^*, \quad \text{Cap} \subseteq A^*, \\ \text{Domain} &= \text{WitnessDomain}, \quad \text{Order} = \{<, =, >, \perp\}, \\ \text{Prov} &= (\text{src}, \text{rev}, \text{cmd}, \text{seed}, \text{host}), \\ \text{Limit} &= \text{NonClaim}_{\text{Rings}} \cup \text{Obligation}. \end{aligned}$$

$$\begin{aligned} \text{NonClaim}_{\text{Rings}} &= \{\text{Novel}(\text{DHT}), \text{Novel}(\text{DID})\} \\ &\quad \cup \{\text{Novel}(\text{WebRTC})\} \\ &\quad \cup \{\text{Novel}(\text{CRDT}), \text{Novel}(\text{Crypto})\} \\ &\quad \cup \{\text{ByzLive}, \text{Consensus}, \text{WANPerf}\}. \end{aligned}$$

Definition 3 (Contribution boundary).

$$\begin{aligned} \text{Reuse}_{\text{Rings}} &= \{\text{Chord}, \text{CorrectChord}, \text{DID}, \text{WebRTC}, \text{CRDT}\}, \\ \text{New}_{\text{Rings}} &= (\Omega_N, \text{PMor}, \text{CarrierSep}, \text{LedgerDisc}_{\text{Rings}}). \end{aligned}$$

$$\begin{aligned} \text{Contribution}_{\text{Rings}} &\equiv \text{Reuse}_{\text{Rings}} \oplus \text{New}_{\text{Rings}} \\ &\models \text{Core}_{\text{Rings}} \wedge \text{LedgerDisc}_{\text{Rings}}. \end{aligned}$$

The formal theorem is Theorem 1; $\text{LedgerDisc}_{\text{Rings}}$ is the admission discipline, not a performance theorem.

Claim family	Ledger class	Carrier in this paper
Ω_N, PMor	Spec	Definitions and Theorem 1
Chord state	$\text{Spec} \cup \text{Ev}_\Gamma$	Algorithms 1–3; finite checks
DID/session	Impl	signed session and context predicates
Affine storage	$\text{Impl} \cup \text{Ev}_\Gamma$	replica-key invariant and storage tests
E2E frames	$\text{Impl} \cup \text{Ev}_\Gamma$	signed frame and receiver policy
Chunking	$\text{Impl} \cup \text{Ev}_\Gamma$	Algorithm 6; bounded re-assembly
Deferred APIs	Spec $\cup$ Obligation	mailbox, hidden descriptor, secret sharing
Wit/rank/sec.	Spec $\cup$ Obligation	policy predicates, no global security theorem
Benchmark rows	ReportedOnly	baseline/runtime coordinates, not admitted eval

Table I

Claim-ledger instantiation. Rows inside **ClaimKind** may be admitted at the listed classes; **ReportedOnly** rows are coordinates, not claim evidence.

III. Algebraic System Model

Assumption 1 (System and adversary scope).

$$\begin{aligned} \kappa &\in \mathbb{N}, \\ \text{fault}_{\text{overlay}} &= \text{fail-stop}, \\ \text{Adv}_{\text{net}} &= \left\{ \begin{array}{l} \text{delay, drop, dup,} \\ \text{reorder, replay} \end{array} \right\}, \\ \text{Adv}_{\text{id}} &= \left\{ \begin{array}{l} \text{manyDID, stolenSession,} \\ \text{staleProof} \end{array} \right\}, \\ \Pr[\text{Forge}_\Pi(A, \kappa)] &\leq \epsilon_\Pi(\kappa), \\ \Pr[\text{Preimage}_H(A, \kappa)] &\leq \epsilon_H(\kappa), \\ \epsilon_\Pi, \epsilon_H &\in \text{negl}, \\ \text{consensus} &\notin \text{Overlay}, \\ \text{order} &= \text{SidecarWitness}, \\ \text{crypto_ops} \cap \text{route_ops} &= \emptyset. \end{aligned}$$

Thus **CorrectChord** is the safety root only for crash-scoped overlay maintenance. Cryptographic claims are computational assumptions; compromised keys and Byzantine routing are adversarial cases. Eclipse resistance, Sybil resistance, anonymity, and denial-of-service resistance must be discharged by admission, session typing, ranking, relay, quota, and measurement predicates at their own layers.

Definition 4 (Identifier carrier).

$$\begin{aligned} M &= 2^{160}, \\ R &= \mathbb{Z}/M\mathbb{Z}, \\ \delta_a(x) &= (x - a) \bmod M, \\ (a, b)_R &= \{x \in R : 0 < \delta_a(x) \leq \delta_a(b)\}, \\ (a, b)_R &= \{x \in R : 0 < \delta_a(x) < \delta_a(b)\}, \\ \text{RouteOps}_R &= \{0, +, -, \delta_a, (_, _)_R, (_, _)_R\}, \\ \text{Mult}_R &\notin \text{RouteOps}_R. \end{aligned}$$

Definition 5 (Live topology). □

$$\begin{aligned}
N &\subset R, \quad 1 \leq |N| < \infty, \\
\text{owner}_N(k) &= \arg \min_{u \in N} \delta_k(u), \\
\text{cw}_N(n) &= \text{sort}_{\delta_n}(N \setminus \{n\}) \cdot \langle n \rangle, \\
\text{succ}_N(n) &= \text{head}(\text{cw}_N(n)), \\
\text{pred}_N(n) &= \begin{cases} \perp, & |N| = 1, \\ \arg \min_{p \in N \setminus \{n\}} \delta_p(n), & |N| > 1, \end{cases} \\
\text{Succ}_N^q(n) &= \text{take}_{\min(q, |N|)}(\text{cw}_N(n)).
\end{aligned}$$

The list $\text{cw}_N(n)$ contains strictly clockwise live nodes and uses n only as the wrap-around sentinel. Hence a singleton ring has $\text{Succ}_N^q(n) = \langle n \rangle$ and $\text{pred}_N(n) = \perp$.

Assumption 2 (Correct-Chord stable base).

$$\begin{aligned}
r &\in \mathbb{N}_{>0}, \\
\text{Steady}(N, r) &\Rightarrow |N| \geq r + 1, \\
\text{Init}(N, r) &\Leftrightarrow 1 \leq |N| \leq r, \\
|\text{Succ}_N^r(n)| &= \min(r, |N|), \\
\text{Maintain} &= \{\text{join}, \text{rectify}\} \\
&\quad \cup \{\text{stabilize}\}_{\text{CorrectChord}}.
\end{aligned}$$

CorrectChord claims in this paper are stable-ring claims under $\text{Steady}(N, r)$; singleton and small-ring behavior is an explicit initialization edge case, not evidence for r distinct live successors.

Definition 6 (Affine replica set).

$$\begin{aligned}
1 \leq \rho &\leq M, \\
\text{repkey}_i^\rho(k) &= (k + \lfloor M \cdot i / \rho \rfloor) \bmod M, \quad 0 \leq i < \rho, \\
\text{RepKey}^\rho(k) &= \langle \text{repkey}_i^\rho(k) : 0 \leq i < \rho \rangle, \\
\text{RepOwner}_N^\rho(k) &= \langle \text{owner}_N(\text{repkey}_i^\rho(k)) : 0 \leq i < \rho \rangle. \\
P_k &= \text{set}(\text{RepKey}^\rho(k)), \\
O_k &= \text{set}(\text{RepOwner}_N^\rho(k)), \\
|P_k| &= \rho, \\
1 \leq |O_k| &\leq \min(\rho, |N|), \\
\text{collapse}_{N, \rho}(k) &\Leftrightarrow |O_k| < \min(\rho, |N|).
\end{aligned}$$

Replica keys specify intended placement cardinality; live owner images specify realizable redundancy under the current live topology. CorrectChord successor lists remain the topology safety root.

Definition 7 (Rings overlay object).

$$\begin{aligned}
\mathcal{R}_N &= (R, N, \text{owner}_N, \text{pred}_N, \text{Succ}_N^r, \text{RepOwner}_N^\rho, F, E), \\
F &\subseteq N \times \mathbb{N} \times N, \\
E &= \prod_{\nu \in \text{NS}} E_\nu, \\
\text{SafetyRoot}(\mathcal{R}_N) &= (\text{pred}_N, \text{Succ}_N^r), \\
\text{PerfWitness}(\mathcal{R}_N) &= F.
\end{aligned}$$

Proposition 1 (Rotation equivariance).

$$\begin{aligned}
t \in R, \quad N + t &= \{n + t : n \in N\} \Rightarrow \\
\text{owner}_{N+t}(k + t) &= \text{owner}_N(k) + t.
\end{aligned}$$

Proof.

$$\begin{aligned}
\forall n \in N. \quad \delta_{k+t}(n + t) &= ((n + t) - (k + t)) \bmod M \\
&= \delta_k(n).
\end{aligned}$$

Constraint 1 (Carrier separation).

$$\begin{aligned}
\text{PlacementOps} &= \{0, +, -, \delta_a, (a, b]_R, (a, b)_R\} \\
&\quad \cup \{\text{owner}_N\} \\
&\quad \cup \{\text{RepKey}^\rho\} \\
&\quad \cup \{\text{RepOwner}_N^\rho\}, \\
\text{CryptoOps} &= \{\times, ^{-1}, \cdot_{\text{scalar}}, \text{interp}\} \\
&\quad \cup \{\text{pair}, \text{subgroup}\}, \\
\text{RunOps} &= \text{TransportOps} \cup \text{EffectOps}, \\
\text{PlacementOps} \cap \text{CryptoOps} &= \emptyset, \\
\text{PlacementOps} \cap \text{RunOps} &= \emptyset, \\
\text{CryptoCarrier} &\in \{\mathbb{F}_q, \mathbb{G}, \mathbb{Z}_q\}, \\
\pi_R &: \text{Ops} \rightarrow \text{End}(R), \\
a \in \text{CryptoOps} \cup \text{RunOps} &\Rightarrow \begin{cases} \pi_R(a) = \text{id}_R, \\ a \notin \text{PlacementOps}. \end{cases}
\end{aligned}$$

Object	Carrier or law
Chord IDs	additive cyclic group R
DID proof	signature verifier and transcript law
VID	$H_\nu : X_\nu \rightarrow R$ then owner map
placement	
Storage	join semilattice or explicit finalizer
merge	
E2E encryption	group operation plus byte-to-point adapter
Sequencing	partial order over witness domains
Effects	writer monad over action lists

Table II

Carrier boundaries in the Rings model

Table II is the carrier-level summary used by the separation constraint; it is not an additional security or performance claim.

IV. Architecture As Interfaces

Definition 8 (Layer tuple).

$$\begin{aligned}
\mathcal{A} &= (I, T, O, X, P, L) \\
I &\mid (DID, \Pi, \text{Session}) \\
T &\mid (C, A_T, \text{offer}, \text{answer}, \text{send}, \text{recv}) \\
O &\mid \mathcal{R}_N \\
X &\mid (S_X, A_X, \text{cap}, I_X) \\
P &\mid \{\mathcal{P}_\nu\}_{\nu \in \text{NS}} \\
L &\mid \text{App} \circ P
\end{aligned}$$

The interface tuple $\mathcal{A}$ separates identity, transport, overlay, extension runtime, protocol, and application layers. Only the overlay owns $\mathcal{R}_N$; the other layers discharge typed obligations through I, T, X, P .

Definition 9 (Identity proof).

$$\begin{aligned}
\Pi &= (K, \Sigma, V, \text{did}, \text{enc}, \text{ttl}) \\
V &: K \times \Sigma \times B^* \rightarrow \{\top, \perp\}, \\
\text{did} &: K \rightarrow X_D, \\
\text{enc} &: \Sigma \rightarrow B^*, \\
\text{ttl} &: \Sigma \rightarrow \text{Time}.
\end{aligned}$$

Invariant 1 (Typed session authority).

$$\text{sid} = (d, s, e, \text{ctx}, \text{cap}), \quad d \in X_D, \quad s \in K, \quad \text{now} < e.$$

$$\text{Session}(d, s, e, ctx, cap) \Leftrightarrow V(k_d, \sigma_{d \rightarrow s}, d \| s \| e \| ctx \| cap) = \top \\ \wedge \text{KeyOf}(d) = k_d \wedge \text{did}(k_d) = d.$$

$$\text{HopCheck}(sid) \equiv \text{Session}(d, s, e, ctx, cap),$$

$$\text{BoundCtx}(sid) = (sid, \nu, path, dst),$$

$$\text{FrameCtx}(f) = (sid, \nu, path, dst),$$

$$\text{Accept}_\Delta(f) \Rightarrow \text{SessionValid}(sid)$$

$$\wedge \text{FrameCtx}(f) = \text{BoundCtx}(sid).$$

Definition 10 (Transport interface).

$$T = (C, \text{offer}, \text{answer}, \text{send}, \text{recv})$$

$$A_T = \{\text{Offer}, \text{Answer}, \text{Ice}, \text{Open}, \text{Send}, \text{Recv}, \text{Close}\}.$$

$$I_T : A_T^* \rightarrow \text{IO}_{\text{WebRTC}} \cup \text{IO}_{\text{Native}}.$$

WebRTC signaling is a runtime adapter rather than a new overlay primitive. Rings may batch offer, answer, and ICE material into fewer protocol actions, but correctness remains delegated to the WebRTC/ICE state machines; evaluation therefore reports connection setup separately from overlay lookup.

Definition 11 (Runtime interpreter).

$$\mathcal{X} = (S_X, A_X, \text{cap}, I_X)$$

$$\text{cap} \subseteq A_X,$$

$$I_X : A_X^* \rightarrow \text{IO},$$

$$I_X(\epsilon) = \text{Noop},$$

$$I_X(\alpha \cdot \beta) = I_X(\alpha); I_X(\beta).$$

WebAssembly is modeled only through I_X and cap : it interprets effect lists under runtime capabilities and does not change the overlay object.

Obligation 2 (Runtime confinement).

$$\tau : S \times I \rightarrow \mathcal{E} + (S \times O \times A_X^*), \\ \text{emit}(\tau(s, i)) \subseteq \text{cap}, \\ a \notin \text{cap} \Rightarrow I_X(a) = \text{Reject}(a).$$

V. Category and Effects

Definition 12 (Namespace object).

$$\nu \in \mathbf{NS}, \\ H_\nu : X_\nu \rightarrow R, \\ (X_\nu, H_\nu) \in \mathbf{Set}/R.$$

Definition 13 (Placement functor).

$$\Omega_N : \mathbf{Set}/R \rightarrow \mathbf{Set}/N, \\ (X, H) \mapsto (X, \text{owner}_N \circ H).$$

$$P_\nu^N = \text{owner}_N \circ H_\nu : X_\nu \rightarrow N.$$

Proposition 2 (Placement functoriality).

$$f : X_\nu \rightarrow X_\mu, \quad H_\mu \circ f = H_\nu.$$

$$P_\mu^N \circ f = P_\nu^N.$$

Proof.

$$\text{owner}_N \circ H_\mu \circ f = \text{owner}_N \circ H_\nu.$$

Definition 14 (Writer effect monad).

$$A^* = \text{FreeMonoid}(A), \\ \mathsf{T}_A(X) = X \times A^*, \\ \eta(x) = (x, \epsilon), \\ (x, \alpha) \gg= f = \text{let } f(x) = (y, \beta) \text{ in } (y, \alpha \cdot \beta).$$

Proposition 3 (Effect associativity).

$$(h^* \circ g^*) \circ f^* = h^* \circ (g^* \circ f^*).$$

Proof.

$$((\alpha \cdot \beta) \cdot \gamma) = (\alpha \cdot (\beta \cdot \gamma)).$$

□

Definition 15 (Typed failure stack).

$$\mathsf{T}_A^\mathcal{E}(X) = \mathcal{E} + (X \times A^*).$$

$$\pi_A(e \in \mathcal{E}) = \epsilon,$$

$$\pi_A(x, \alpha) = \alpha,$$

$$\text{RecoverableReport} \in A.$$

Definition 16 (Protocol object).

$$\mathcal{P}_\nu = (X_\nu, S_\nu, I_\nu, O_\nu, A_\nu, \mathcal{E}_\nu, H_\nu, \tau_\nu)$$

$$\tau_\nu : S_\nu \times I_\nu \rightarrow \mathsf{T}_{A_\nu}^\mathcal{E}(S_\nu \times O_\nu).$$

Definition 17 (Protocol morphism).

$$\Sigma : \mathcal{P}_\nu \rightarrow \mathcal{P}_\mu = (\sigma_X, \sigma_S, \sigma_I, \sigma_O, \sigma_\mathcal{E}, \sigma_A, \sigma_A^*)$$

$$\sigma_X : X_\nu \rightarrow X_\mu, \quad \sigma_S : S_\nu \rightarrow S_\mu,$$

$$\sigma_I : I_\nu \rightarrow I_\mu, \quad \sigma_O : O_\nu \rightarrow O_\mu,$$

$$\sigma_\mathcal{E} : \mathcal{E}_\nu \rightarrow \mathcal{E}_\mu, \quad \sigma_A : A_\nu \rightarrow A_\mu,$$

$$\sigma_A^* : A_\nu^* \rightarrow A_\mu^*,$$

$$\sigma_A^*(\epsilon) = \epsilon,$$

$$\sigma_A^*(a \cdot \alpha) = \sigma_A(a) \cdot \sigma_A^*(\alpha),$$

$$\sigma_A^*(\alpha \cdot \beta) = \sigma_A^*(\alpha) \cdot \sigma_A^*(\beta),$$

$$H_\mu \circ \sigma_X = H_\nu.$$

$$\mathsf{T}_{\sigma_A}^{\sigma_\mathcal{E}}(e) = \sigma_\mathcal{E}(e),$$

$$\mathsf{T}_{\sigma_A}^{\sigma_\mathcal{E}}(s, o, \alpha) = (\sigma_S(s), \sigma_O(o), \sigma_A^*(\alpha)),$$

$$\mathsf{T}_{\sigma_A}^{\sigma_\mathcal{E}} \circ \tau_\nu = \tau_\mu \circ (\sigma_S \times \sigma_I).$$

The predicate $\text{PMor}_{\nu\mu}(\Sigma)$ is the conjunction of these typing and preservation equations.

Lemma 1 (Owner totality). If $1 \leq |N| < \infty$ and $H_\nu : X_\nu \rightarrow R$, then

$$P_\nu^N = \text{owner}_N \circ H_\nu : X_\nu \rightarrow N$$

is total.

Proof.

$$\forall k \in R. \quad \text{owner}_N(k) = \arg \min_{u \in N} \delta_k(u) \in N.$$

Finite nonempty N gives existence; $\delta_k : R \rightarrow \mathbb{N}$ gives a total order key. □

Lemma 2 (Morphism preservation).

$$\text{PMor}_{\nu\mu}(\Sigma_{\nu\mu}) \Rightarrow P_\mu^N \circ \sigma_X = P_\nu^N.$$

Proof.

$$\square \quad P_\mu^N \circ \sigma_X = \text{owner}_N \circ H_\mu \circ \sigma_X = \text{owner}_N \circ H_\nu = P_\nu^N. \quad \square$$

Lemma 3 (Effect confinement).

$$\forall \nu \in \mathbf{NS}. \forall s, i. \quad \tau_\nu(s, i) \in \mathbf{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu).$$

Moreover, if $\text{emit}(\tau_\nu(s, i)) \subseteq \text{cap}$, then

$$I_X(\pi_A(\tau_\nu(s, i))) \in \mathbf{IO} \quad \wedge \quad a \notin \text{cap} \Rightarrow I_X(a) = \text{Reject}(a).$$

Proof. The first formula is the type of τ_ν . The second is exactly the runtime confinement obligation for the interpreter I_X . $\square$

Lemma 4 (Carrier non-interference).

$$\begin{aligned} a &\in \text{CryptoOps} \cup \text{RunOps} \\ \Rightarrow \quad &\forall (X, H) \in \mathbf{Set}/R. \\ &\left\{ \begin{array}{l} \Omega_N(X, \pi_R(a) \circ H) = \Omega_N(X, H), \\ \Omega_N(X, H) = (X, \text{owner}_N \circ H). \end{array} \right. \end{aligned}$$

Proof. By Constraint 1, such a has $\pi_R(a) = \text{id}_R$. Hence $\text{owner}_N \circ \pi_R(a) \circ H = \text{owner}_N \circ H$. $\square$

Theorem 1 (Carrier-separated ownership theorem). Assume $1 \leq |N| < \infty$, CorrectChord stable-base obligations, carrier separation, and protocol morphisms satisfying Definition 17. Then

$$\begin{aligned} \text{OwnerLaw} &\equiv \forall \nu, x. \\ &\quad P_\nu^N(x) = \text{owner}_N(H_\nu(x)) \in N, \\ \text{MorphLaw} &\equiv \forall \nu, \mu, \Sigma. \\ &\quad \text{PMor}_{\nu\mu}(\Sigma) \Rightarrow P_\mu^N \circ \sigma_X = P_\nu^N, \\ \text{EffectLaw} &\equiv \forall \nu, s, i. \\ &\quad \tau_\nu(s, i) \in \mathbf{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu), \\ \text{CarrierLaw} &\equiv \forall a \in \text{CryptoOps} \cup \text{RunOps}. \\ &\quad \forall (X, H) \in \mathbf{Set}/R. \\ &\quad \Omega_N(X, \pi_R(a) \circ H) = \Omega_N(X, H), \\ \text{Core}_{\text{Rings}} &\equiv \text{OwnerLaw} \wedge \text{MorphLaw} \\ &\quad \wedge \text{EffectLaw} \wedge \text{CarrierLaw}. \end{aligned}$$

Proof. OwnerLaw is Lemma 1; MorphLaw is Lemma 2; EffectLaw is Lemma 3; CarrierLaw is Lemma 4. $\square$

Constraint 2 (Claim ledger discipline).

$$\begin{aligned} \text{Admit}_R(c, k) &\Rightarrow k \in \text{ClaimKind} \wedge \text{Support}_R(c, k), \\ \text{Admit}_R(c, \text{Ev}_\Gamma) &\Rightarrow \text{finite}(\Gamma) \wedge \text{declared}(\Gamma), \\ \text{Support}_R(c, \text{Spec}) &\Rightarrow c \in \left\{ \begin{array}{l} \text{OwnerLaw, MorphLaw,} \\ \text{EffectLaw, CarrierLaw} \end{array} \right\}, \\ \text{Support}_R(c, \text{Impl}) &\Rightarrow \text{Anchor}_R(c) \in \text{Manifest}_R, \\ \text{Support}_R(c, \text{Ev}_\Gamma) &\Rightarrow \text{Witness}_R(c, \Gamma), \\ \text{Support}_R(c, \text{Obligation}) &\Rightarrow c \in \text{Obligation}, \\ \text{Reject}_R(c) &\Leftrightarrow c \in \text{NonClaim}_{\text{Rings}}, \\ \text{Reject}_R(c) &\Rightarrow \neg \exists k. \text{Admit}_R(c, k), \\ \text{LedgerDisc}_{\text{Rings}} &= \text{Admit}_R \wedge \text{Reject}_R. \end{aligned}$$

Thus non-claims are excluded by the admission judgment; they are not proved unreachable by Theorem 1.

Proposition 4 (Claim-ledger admissibility).

$$\begin{aligned} \text{Admit}_R(c, k) &\Rightarrow k \in \text{ClaimKind} \wedge \text{Support}_R(c, k), \\ c \in \text{NonClaim}_{\text{Rings}} &\Rightarrow \neg \text{Admit}_R(c), \\ \text{Admit}_R(c, \text{Ev}_\Gamma) &\Rightarrow \left\{ \begin{array}{l} \text{finite}(\Gamma), \\ \text{declared}(\Gamma). \end{array} \right. \end{aligned}$$

Proof. Immediate from Constraint 2 and Table I. $\square$

VI. Chord Overlay

$$\begin{aligned} \text{Base} &= \text{Chord}, \\ \text{Maintenance} &= \text{CorrectChord}, \\ \text{Object} &= \mathcal{R}_N, \\ \text{Witnesses} &= \{\text{Algorithms 1, 2, 3}\}. \end{aligned}$$

This section fixes Chord as the base protocol and CorrectChord as the maintenance law [4], [5].

Definition 18 (Local node state and Chord auxiliaries).

$$\begin{aligned} S_n &= (p_n, L_n, F_n, E_n), \\ n &\in N \subset R, \\ p_n &\in N \cup \{\perp\}, \quad L_n \in N^{\leq r}, \\ F_n &: \{0, \dots, 159\} \rightarrow N. \end{aligned}$$

The algorithms are written over identifiers $n \in N$; an implementation node record $\hat{n}$ is projected by $\text{id}(\hat{n}) = n$.

$$\begin{aligned} \text{head}(\langle \rangle) &= \perp, \\ \text{Normalize}_r(n, X) &= \text{take}_{\min(r, |N|)}(U_n(X)), \\ U_n(X) &= \text{uniq}(\text{filter}_N(X) \cdot \text{cw}_N(n)), \\ \text{target}_i(n) &= n + 2^i \bmod M, \\ \text{finger}_i(n) &= \text{owner}_N(\text{target}_i(n)), \\ \text{FingerInv}(N, S) &\Leftrightarrow \forall n \in N. \forall i < 160. \\ &\quad \left\{ \begin{array}{l} F_n(i) \downarrow, \\ F_n(i) = \text{finger}_i(n). \end{array} \right. , \\ \text{RouteSet}_S(n) &= \text{rng}(F_n) \cup \text{set}(L_n), \\ \text{Cand}_S(n, k) &= \text{RouteSet}_S(n) \cap (n, k)_R, \\ \text{ClosestPreceding}_S(n, k) &= \max_{\delta_n} \text{Cand}_S(n, k). \end{aligned}$$

Here $\text{ClosestPreceding}_S(n, k) = \perp$ when the displayed set is empty.

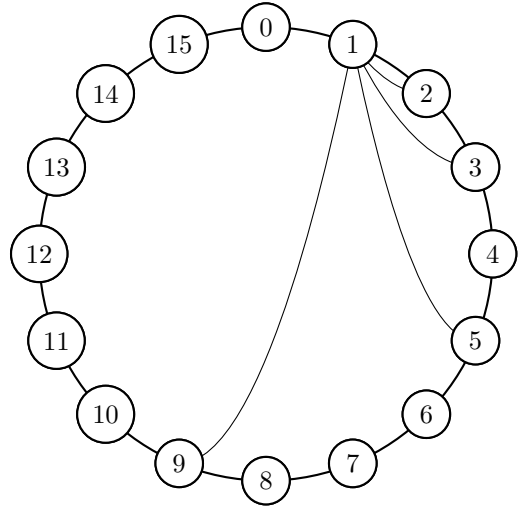


Figure 1. Chord finger targets from node 1 in a 16-node ring

Figure 1 illustrates finger targets only; lookup correctness is the interval invariant below.

Algorithm 4 fixes the notation for the biased circular order used by placement comparisons.

Algorithm 1 DHT Join and Rectify Transitions

```

1: function BeginJoin( $n, known$ )
2:    $pred[n] \leftarrow \perp$ 
3:   if  $known = \perp$  then
4:      $succ[n] \leftarrow \langle n \rangle$ 
5:     return Done
6:   return Remote( $known, FindSuccessor(n)$ )
7: function InstallSuccessor( $n, s$ )
8:   if  $s = \perp$  then
9:     return Repair(DiscoverSuccessor( $n$ ))
10:   $succ[n] \leftarrow \text{Normalize}_r(n, \langle s \rangle)$ 
11:  return Remote( $s, QuerySuccList$ )
12: function InstallSuccList( $n, s, list$ )
13:   $succ[n] \leftarrow \text{Normalize}_r(n, \langle s \rangle \cdot list)$ 
14:   $h \leftarrow head(succ[n])$ 
15:  if  $h = \perp$  then
16:    return Repair(DiscoverSuccessor( $n$ ))
17:  else if  $h = n$  then
18:    return Done
19:  else
20:     $a \leftarrow \text{Remote}(h, Notify(n))$ 
21:     $b \leftarrow \text{Remote}(h, SyncStorage(n))$ 
22:    return Multi( $a, b$ )
23: function HandleNotify( $n, q$ )
24:   if  $q \neq n \wedge (pred[n] = \perp \vee q \in (pred[n], n)_R)$  then
25:      $pred[n] \leftarrow q$ 
26:   return Noop

```

Algorithm 2 DHT Lookup Algorithm

```

1: function FindSuccessor( $key, n$ )
2:   if  $key = n$  then
3:     return Found( $n$ )
4:    $s \leftarrow head(succ[n])$ 
5:   if  $s = \perp$  then
6:     return Repair(QuerySuccList( $n$ ))
7:   else if  $s = n$  then
8:     return Found( $n$ )
9:   else if  $key \in (n, s]_R$  then
10:    return Found( $s$ )
11:   $next \leftarrow \text{ClosestPreceding}_S(n, key)$ 
12:  if  $next = \perp$  or  $next = n$  then
13:     $next \leftarrow s$ 
14:  return Remote( $next, FindSuccessor(key)$ )

```

Invariant 2 (Ring fixpoint).

$$\begin{aligned}
\text{Inv}_{ring}(N, S) \equiv & \forall n \in N. p_n = \text{pred}_N(n) \\
& \wedge L_n = \text{Succ}_N^r(n) \\
& \wedge \text{head}(L_n) = \text{succ}_N(n).
\end{aligned}$$

Invariant 3 (Lookup owner preservation).

$$\text{Eval}_S(n, k) \Downarrow o$$

be the fair closure of Algorithm 2, where *Remote*($m, FindSuccessor(k)$) resumes the same lookup

Algorithm 3 DHT Stabilization

```

1: function BeginStabilize( $n$ )
2:    $s \leftarrow head(succ[n])$ 
3:   if  $s = \perp$  then
4:     return Repair(DiscoverSuccessor( $n$ ))
5:   else if  $s = n$  then
6:     return Noop
7:   else
8:     return Remote( $s, QueryTopology$ )
9: procedure Stabilize( $n, topo$ )
10:   $actions \leftarrow []$ 
11:   $s \leftarrow head(succ[n])$ 
12:  if  $s = \perp$  then
13:    return Repair(DiscoverSuccessor( $n$ ))
14:  else if  $s = n$  then
15:    return Noop
16:   $c \leftarrow topo.pred$ 
17:  if  $c \neq \perp$  and  $c \neq n$  and  $c \in (n, s)_R$  then
18:     $succ[n] \leftarrow \text{Normalize}_r(n, \langle c \rangle \cdot succ[n])$ 
19:   $succ[n] \leftarrow \text{Normalize}_r(n, succ[n] \cdot topo.succ)$ 
20:   $h \leftarrow head(succ[n])$ 
21:  if  $h \neq \perp$  and  $h \neq n$  then
22:     $a \leftarrow \text{Remote}(h, Notify(n))$ 
23:     $b \leftarrow \text{Remote}(h, QuerySuccList)$ 
24:     $actions \leftarrow actions \cdot [a, b]$ 
25:  return Multi( $actions$ )

```

at node m . Then

$$\begin{aligned}
& (\text{Steady}(N, r) \vee |N| = 1) \wedge \text{Inv}_{ring}(N, S) \\
& \Rightarrow \forall n \in N. \forall k \in R. \text{Eval}_S(n, k) \Downarrow \text{owner}_N(k).
\end{aligned}$$

Proof obligation. If Algorithm 2 returns *Found*(s) by the interval test, then $s = head(L_n) = \text{succ}_N(n)$ and no live node lies in $(n, s)_R$; therefore $\text{owner}_N(k) = s$. If it forwards, the chosen target is either the closest preceding finger/list entry or the first successor; in both cases it lies in $(n, k)_R$, so $\delta_{next}(k) < \delta_n(k)$. Finiteness of N gives termination. The branches $key = n$ and $s = n$ discharge the singleton and exact-owner edge cases. Repair branches are outside the steady-ring premise. $\square$

VII. DID and Resource Algebra

Definition 19 (DID namespace).

$$\mathcal{D} = (X_D, H_D, \Pi, \text{Session}), \quad H_D : X_D \rightarrow R.$$

$$\begin{aligned}
D_K(k) &= H_D(\text{did}(k)), \\
\text{owner}_N(D_K(k)) &\in N, \\
\text{Session} &\subseteq X_D \times K \times \text{Time} \times \text{Ctx} \times \text{Cap}.
\end{aligned}$$

Definition 20 (VID namespace).

$$\begin{aligned}
\mathcal{V} &= (X_V, H_V, \text{Auth}_V), \\
H_V &: X_V \rightarrow R, \\
\text{PrivateKey}(H_V(x)) &= \perp, \\
\text{Auth}_V(op, x, \sigma, caps) &\in \{\top, \perp\}.
\end{aligned}$$

Definition 21 (Biased order).

$$A \leq_C B \iff \delta_C(A) \leq \delta_C(B).$$

Algorithm 4 Biased Circular Comparison

```

1: function BiasedCompare( $A, B, C$ )
2:    $a \leftarrow \delta_C(A)$ 
3:    $b \leftarrow \delta_C(B)$ 
4:   if  $a < b$  then
5:     return  $A <_C B$ 
6:   else if  $a > b$  then
7:     return  $B <_C A$ 
8:   else
9:     return  $A = B$ 

```

Definition 22 (Replicated entry algebra).

$$\begin{aligned}
\mathcal{E}_v^{join} &= (E_v, \sqsubseteq, \sqcup, \perp), \\
\mathcal{E}_v^{final} &= (E_v, \text{conflict}_v, \text{finalize}_v). \\
x \sqcup y &= y \sqcup x, \\
(x \sqcup y) \sqcup z &= x \sqcup (y \sqcup z), \\
x \sqcup x &= x, \\
x \sqcup \perp &= x, \\
\text{finalize}_v &: \text{conflict}_v(E_v) \rightarrow E_v. \\
\text{case}_1 &= \text{CRDT}.
\end{aligned}$$

This is the CRDT join-semilattice case [6].

Invariant 4 (Placement).

$$\begin{aligned}
P_{\nu,x} &= \text{set}(\text{RepKey}^\rho(H_\nu(x))), \\
K_{\nu,x}^{impl} &\subseteq R, \\
\text{Inv}_{place} &\equiv \forall \nu, x. K_{\nu,x}^{impl} \subseteq P_{\nu,x}.
\end{aligned}$$

Obligation 3 (Replica realization).

$$\begin{aligned}
O_{\nu,x} &= \text{set}(\text{RepOwner}_N^\rho(H_\nu(x))), \\
L_{\nu,x} &= |O_{\nu,x}|, \\
\text{FullRed}_{\nu,x} &\Leftrightarrow L_{\nu,x} = \min(\rho, |N|), \\
\text{Collapse}_{\nu,x} &\Leftrightarrow L_{\nu,x} < \min(\rho, |N|), \\
\text{LiveClaim}_{\nu,x}(\rho) &\Leftrightarrow \text{Admit}_R(\text{FullRed}_{\nu,x}, \text{Evidence}_R), \\
\text{LiveClaim}_{\nu,x}(\rho) &\Rightarrow \text{Inv}_{place} \wedge \text{FullRed}_{\nu,x}, \\
\text{ReplicaGap}_R &= \{(\nu, x) : \text{Collapse}_{\nu,x}\}, \\
\text{ReplicaGap}_R \neq \emptyset &\Rightarrow \text{fullReplicaRealization} \in \text{Obligation}.
\end{aligned}$$

When $|N| < \rho$, fewer than ρ distinct live owners is not collapse; collapse means fewer owners than the live topology can realize.

Definition 23 (Mailbox).

$$\begin{aligned}
\text{Class}(\text{Mailbox}) &\subseteq \{\text{Spec}, \text{Obligation}\}. \\
\text{mbox}(d, e) &= H_M(\text{rings.mailbox} \| d \| e), \\
\text{entry}_M &= (\eta, \text{exp}, \text{proof}_s, \text{dst}, \text{cipher}).
\end{aligned}$$

Definition 24 (Hidden service descriptor).

$$\begin{aligned}
\text{Class}(\text{HiddenDesc}) &\subseteq \{\text{Spec}, \text{Obligation}\}. \\
h &= (\nu, \rho, \text{Intro}, \text{Relay}, \\
&\quad \text{Cap}, K, \text{exp}, \text{rank}), \\
\text{vid}(h) &= H_H(h), \\
\text{Lookup}(h) &= P_H^N(h), \\
\text{Valid}(h, \sigma) &= \text{Auth}_H(h, \sigma).
\end{aligned}$$

Definition 25 (Subring).

$$\begin{aligned}
\text{Class}(\text{Subring}) &= \text{Impl}. \\
\mathcal{S}_u &= (u, N_u, \text{owner}_{N_u}, \text{Succ}_{N_u}^{r_u}, E_u) \\
u &= H_U(\text{policy}), \\
\text{memberlog}_u &\in \text{MergeLog} \cup \text{FinalState}.
\end{aligned}$$

VIII. Cryptographic Carriers and Traffic

Definition 26 (End-to-end carrier).

$$\begin{aligned}
|\mathbb{G}| &= q, \\
\text{Scalar}(\mathbb{G}) &= \mathbb{Z}_q, \\
x &\in \mathbb{Z}_q, \\
h &= g^x, \\
\mathbb{G} &\neq R.
\end{aligned}$$

Algorithm 5 ElGamal Encryption and Decryption

```

1: Input: Encoded group element  $m_G \in \mathbb{G}$ , private key
    $x \in \mathbb{Z}_q$ , public key  $h = g^x$ , generator  $g$ 
2: Output: Group ciphertext  $(c_1, c_2)$ 
3: procedure Encryption
4:   Choose a random integer  $k \in \mathbb{Z}_q$ 
5:   Compute  $c_1 = g^k$ 
6:   Compute  $c_2 = m_G \cdot h^k$ 
7:   return  $(c_1, c_2)$ 
8: procedure Decryption
9:   Compute  $m_G = c_2 \cdot (c_1)^{-x}$ 
10:  return  $m_G$ 

```

Algorithm 5 is the group-level carrier operation. Byte payloads must first be encoded into curve points and split into bounded frames. The direct ElGamal body is malleable when detached from its signed transport envelope; Rings therefore treats message signatures, DID ownership of the public key, stream identifiers, sequence numbers, and final-frame checks as part of the security boundary rather than as optional metadata.

Definition 27 (Signed encrypted frame).

$$f = (\text{sid}, \text{seq}, \text{last}, \text{cipher}, \text{pk}_s, \sigma_s).$$

$$\begin{aligned}
\text{sid} &= (d, s, e, \text{ctx}, \text{cap}), \quad d \in X_D, \quad s \in K, \\
m_f &= \text{sid} \| \text{seq} \| \text{last} \| \text{cipher},
\end{aligned}$$

$$\begin{aligned}
\text{FrameValid}(f) &\Leftrightarrow V(\text{pk}_s, \sigma_s, m_f) = \top \\
&\quad \wedge \text{Session}(d, s, e, \text{ctx}, \text{cap}) \\
&\quad \wedge \text{pk}_s = s \\
&\quad \wedge \text{FrameCtx}(f) = \text{BoundCtx}(\text{sid}).
\end{aligned}$$

Definition 28 (Receiver sequence policy).

$$\Delta = (\text{next}, \text{final}, \text{closed}, \text{pending}, \text{seen}_{\text{seq}}, \text{seen}_{\text{fid}}, w).$$

$$\begin{aligned}
\text{pending} &: \mathbb{N} \rightarrow \text{Frame}, \\
\text{NoFuture}_\Delta(\text{seq}) &\Leftrightarrow \neg \exists j \in \text{dom}(\text{pending}). j > \text{seq}, \\
\text{Dup}_\Delta(f) &\Leftrightarrow \text{seq}(f) \in \text{seen}_{\text{seq}} \\
&\quad \vee \text{fid}(f) \in \text{seen}_{\text{fid}} \\
&\quad \vee \text{pending}(\text{seq}(f)) = f, \\
\text{Conflict}_\Delta(f) &\Leftrightarrow \text{seq}(f) \in \text{dom}(\text{pending}) \\
&\quad \wedge \text{pending}(\text{seq}(f)) \neq f, \\
\text{New}_\Delta(f) &\Leftrightarrow \text{seq}(f) \notin \text{seen}_{\text{seq}} \cup \text{dom}(\text{pending}) \\
&\quad \wedge \text{fid}(f) \notin \text{seen}_{\text{fid}} \\
&\quad \wedge \neg \text{closed} \\
&\quad \wedge \text{seq}(f) - \text{next} \leq w \\
&\quad \wedge (\text{final} = \perp \vee \text{seq}(f) \leq \text{final}) \\
&\quad \wedge (\text{last}(f) \Rightarrow \text{NoFuture}_\Delta(\text{seq}(f))), \\
\text{Accept}_\Delta(f) &\Leftrightarrow \text{FrameValid}(f) \\
&\quad \wedge \text{New}_\Delta(f), \\
\text{Reject}_\Delta(f) &\Leftarrow \text{FrameValid}(f) \\
&\quad \wedge \text{Conflict}_\Delta(f), \\
\text{DropDup}_\Delta(f) &\Leftrightarrow \text{FrameValid}(f) \\
&\quad \wedge \text{Dup}_\Delta(f) \\
&\quad \wedge \text{emit} = \epsilon.
\end{aligned}$$

Here w is receiver policy, not a signed wire field. A byte-identical duplicate may be valid and dropped; a same-sequence different pending frame is rejected as equivocation.

Definition 29 (Secret sharing placement).

$$\text{Class}(\text{SecretSharing}) \subseteq \{\text{Spec}, \text{Obligation}\}.$$

$$\begin{aligned}
P &\in \mathbb{F}_q[X], \\
sh_i &= (i, P(i)) \in \mathbb{F}_q^2, \\
H_S(sh_i) &\in R, \\
\text{reconstruct}(\{sh_i\}) &= P(0) \quad (\mathbb{F}_q\text{-interp.})
\end{aligned}$$

This is Shamir reconstruction over the separate field carrier [7].

Definition 30 (Relay frame).

$$\begin{aligned}
r &= (\text{path}, \text{dst}, \nu, \text{session}, \text{payload}, \text{report}). \\
\text{RelayValid}(r) &= \text{SessionValid}(\text{session}) \\
&\quad \wedge \text{dst} \in R, \\
\text{SplitVocabulary} &= \text{MSRP}.
\end{aligned}$$

The split vocabulary follows MSRP [8].

Algorithm 6 End-to-End Chunking Algorithm

```

1: procedure E2E Chunking
2:   Input: data, chunk size
3:   Output: chunks
4:   chunks  $\leftarrow []$ 
5:   total-chunks  $\leftarrow \text{ceil}(\text{len}(\text{data})/\text{chunk-size})$ 
6:   for i in range(total-chunks) do
7:     chunk  $\leftarrow \text{data}[i*\text{chunk-size} : (i+1)*\text{chunk-size}]$ 
8:     append chunk to chunks
9:   return chunks

```

Algorithm 6 gives the bounded split/reassembly shape; it does not claim MSRP interruption or interleaving semantics.

Invariant 5 (Chunk reassembly).

$$\begin{aligned}
\text{complete}(m) &\Leftrightarrow \forall i < \text{total}(m). \exists! c_i \\
&\quad \wedge \sum_i |c_i| \leq B_m \\
&\quad \wedge \text{now} < \text{ttl}(m).
\end{aligned}$$

IX. Ordering, Ranking, and Security

Definition 31 (Sidecar witness).

$$\text{Class}(\text{SidecarWitness}) \subseteq \{\text{Spec}, \text{Obligation}\}.$$

$$w = (\text{domain}, \text{root}, \text{height}, \text{time}, \text{finality}).$$

$$c = H(\text{msg}),$$

$$\begin{aligned}
\text{Witness}(\text{msg}, w) &\Leftrightarrow \text{Include}(c, w.\text{root}) = \top \\
&\quad \wedge \text{Final}(w) = \top.
\end{aligned}$$

Definition 32 (Witness order).

$$\begin{aligned}
a < b &\Leftrightarrow \text{domain}(w_a) = \text{domain}(w_b) \\
&\quad \wedge \text{time}(w_a) < \text{time}(w_b).
\end{aligned}$$

$$\chi : \text{Domain} \times \text{Domain} \rightarrow \text{Order}.$$

Definition 33 (Ranking event).

$$\text{Class}(\text{Ranking}) \subseteq \{\text{Spec}, \text{Obligation}\}.$$

$$\begin{aligned}
q &= (\text{subject}, \text{rater}, \text{behavior}, \\
&\quad \text{evidence}, \text{epoch}, \text{weight}, \sigma).
\end{aligned}$$

$$m_q = \text{subject} \parallel \text{behavior} \parallel \text{evidence} \parallel \text{epoch},$$

$$\text{RankValid}(q) \Leftrightarrow V(k_{\text{rater}}, \sigma, m_q) = \top.$$

Definition 34 (Score aggregation).

$$\begin{aligned}
\text{score}_e(s) &= \text{robust}\{\text{weight}(q) \cdot \text{value}(q) : \\
&\quad q.\text{subject} = s\}.
\end{aligned}$$

$$\text{robust} \in \{\text{median}, \text{trimmedMean}, \text{declared}\}.$$

Attack	Predicate	Boundary claimed here
Sybil	one actor controls many DIDs	not solved by overlay; admission, ranking, quota
Eclipse	biased neighbor or successor view	not claimed under Byzantine routing; requires secure-routing policy
Replay	stale valid signature or frame	typed session, expiry, nonce/sequence set
MITM	wrong key for DID or session	DID proof, signed envelope, context binding
DoS	unbounded storage, relay, or chunks	quotas, TTL, chunk bounds, backpressure
Privacy	route or relay observation	relay policy only; no anonymity theorem

Table III
Threat predicates and claim boundaries

Table III fixes the negative boundary: listed attacks are not discharged by the overlay theorem.

Definition 35 (Security and policy boundary).

$$\begin{aligned}
\text{AllowedRelay} &\subseteq \text{Path}, \\
\text{use, limit} &: \text{User} \times \text{Epoch} \rightarrow \mathbb{N}, \\
\text{SecEvent} &\subseteq X_D \times \text{Epoch} \times \text{RankEvent} \\
&\quad \times \text{RelayFrame} \times \text{User}, \\
P_{adm} &: X_D \times \text{Epoch} \rightarrow \{\top, \perp\}, \\
P_{rank}(q) &\Leftrightarrow \text{RankValid}(q) \wedge \text{weight}(q) \leq W_{max}, \\
P_{relay}(r) &\Leftrightarrow \text{RelayValid}(r) \wedge \text{path}(r) \in \text{AllowedRelay}, \\
P_{quota}(u, e) &\Leftrightarrow \text{use}(u, e) \leq \text{limit}(u, e), \\
z &= (d, e, q, r, u), \\
\text{Policy}_R(z) &\Leftrightarrow P_{adm}(d, e) \\
&\quad \wedge P_{rank}(q) \\
&\quad \wedge P_{relay}(r) \\
&\quad \wedge P_{quota}(u, e). \\
\text{Base} &\not\models \text{SybilResist} \wedge \text{EclipseResist} \\
&\quad \wedge \text{Anon} \wedge \text{DoSResist}, \\
\text{SecClaim}_{core} &\Rightarrow \text{SelfCertID} \wedge \text{TypedSession} \\
&\quad \wedge \text{ReplayReject}, \\
\text{SecClaim}_{policy} &\Rightarrow \forall z \in \text{SecEvent}. \text{Policy}_R(z), \\
\text{SecClaim}_{core} &\not\Rightarrow \text{SecClaim}_{policy}.
\end{aligned}$$

Obligation 4 (Replay and session-confusion rejection).

$$\begin{aligned}
\text{Accept}_\Delta(f, t) &\Rightarrow \text{SessionValid}(\text{sid}(f)) \\
&\quad \wedge t < \text{exp}(\text{sid}(f)) \\
&\quad \wedge \text{ctx}(f) = \text{ctx}(\text{sid}(f)) \\
&\quad \wedge \text{fid}(f) \notin \text{seen}_{fid} \\
&\quad \wedge \text{seq}(f) \notin \text{seen}_{seq}, \\
\text{seen}'_{fid} &= \text{seen}_{fid} \cup \{\text{fid}(f)\}, \\
\text{seen}'_{seq} &= \text{seen}_{seq} \cup \{\text{seq}(f)\}, \\
\text{emit}(f) &\Rightarrow \text{fid}(f) \in \text{seen}'_{fid}, \\
\text{DropDup}_\Delta(f) &\Rightarrow \text{emit} = \epsilon.
\end{aligned}$$

X. Verification Evidence and Evaluation Scope

$$\begin{aligned}
\text{Ledger} &= \text{Spec} \uplus \text{Artifact}_R \\
&\quad \uplus \text{ReportedOnly} \uplus \text{ExtBase}_C \\
&\quad \uplus \text{Obligation}, \\
\text{Artifact}_R &= \text{FiniteModel} \cup \text{TraceReplay} \\
&\quad \cup \text{Tests}_{CI} \\
&\quad \cup \text{FrameImpl} \cup \text{ChunkImpl}, \\
\text{Manifest}_R &= \text{Path} \times \text{Scope} \times \text{Status}, \\
\text{AdmittedArtifact}(a) &\Leftrightarrow a \in \text{Artifact}_R \\
&\quad \wedge \text{Anchor}_R(a) \in \text{Manifest}_R, \\
\text{PerfClaim}_R &\in \text{Obligation}.
\end{aligned}$$

This section separates executable artifact evidence from reported coordinates. The repository contains finite-scope evidence for overlay, storage, and message-flow obligations. It does not yet contain a wide-area Rings benchmark, committed benchmark-table regeneration data, or committed seeds sufficient to regenerate the tables below. Tables IV and V are the artifact ledger; reported comparison coordinates are isolated in Table VI.

$$\begin{aligned}
\text{ProvFull}(d) &\Leftrightarrow \text{src}(d) \neq \perp \wedge \text{rev}(d) \neq \perp \\
&\quad \wedge \text{cmd}(d) \neq \perp \wedge \text{seed}(d) \neq \perp \\
&\quad \wedge \text{host}(d) \neq \perp, \\
\text{AdmittedEval}(d) &\Leftrightarrow \text{Regen}(d) \wedge \text{ProvFull}(d), \\
\neg \text{AdmittedEval}(d) &\Rightarrow d \in \text{ReportedOnly} \cup \text{ExtBase}_C.
\end{aligned}$$

Definition 36 (Execution relation and effect abstraction). Let A_A be concrete scheduled actions and $\mathcal{I}_A$ the abstract inputs of actor A .

$$\begin{aligned}
I_A &: A_A \rightarrow \mathcal{I}_A, \\
I_A^\sharp(\langle a_1, \dots, a_m \rangle) &= \langle I_A(a_j) : I_A(a_j) \downarrow_{j=1}^m \rangle.
\end{aligned}$$

For triples, lift componentwise:

$$\begin{aligned}
I_A^\sharp(s, o, \beta) &= (s, o, I_A^\sharp(\beta)). \\
Q = a :: Q_0 \quad I_A(a) = i \quad \tau_A(S, i) = (S', O, \beta) \\
\hline
(S, Q) &\rightarrow_A (S', Q_0 \cdot \beta)
\end{aligned}$$

$$\begin{aligned}
\text{Explore}_\Gamma &= \text{ModelCheck}(\rightarrow_A, \Gamma), \\
\Gamma &= (|N|, \text{topology}, \text{scheduler}, \text{fault}).
\end{aligned}$$

$$\text{ModelClaim}(\Gamma) \Rightarrow \text{finite}(\Gamma) \wedge \text{declared}(\Gamma).$$

The relation follows TLA+ style and Stateright execution in the declared finite scope [9], [10].

Definition 37 (Effect preorder).

$$\begin{aligned}
(s, o, \alpha) \preceq_A (s', o', \alpha') &\iff s = s' \wedge o = o' \\
&\quad \wedge \text{obs}_A(\alpha) = \text{obs}_A(\alpha').
\end{aligned}$$

Here obs_A erases implementation-local stutter effects such as logging, timers already represented by Γ , and transport bookkeeping not visible in the actor interface.

Invariant 6 (Effect refinement).

$$\forall s, i. \quad I_A^\sharp(\text{Impl}_A(s, i)) \preceq_A I_A^\sharp(\tau_A(s, i)).$$

Equivalently, implementation and specification have equal state/output projections and equal observable abstract effect traces:

$$\pi_{\text{state}, \text{out}}(I_A^\sharp(\text{Impl}_A)) = \pi_{\text{state}, \text{out}}(I_A^\sharp \circ \tau_A).$$

Obligation 5 (Topology coverage).

$$\mathcal{L} = \{\text{line}, \text{wrap}, \text{gap}, \text{partition}, \text{merge}, \text{churn}\}$$

$$\mathcal{S} = \{\text{loss}, \text{dup}, \text{reorder}, \text{timeout}, \text{restart}\}.$$

$$\text{CheckCase} = (L, S, P), \quad L \in \mathcal{L}, \quad S \in \mathcal{S}.$$

Artifact	Checked scope	Non-claim / limit
Topology	join, rectify, stabilize fixpoints	no parametric
tests	predecessor, discovery, CRDT, handoff	TLAPS proof
Stateright	deterministic routing schedules	finite Γ , no live-network theorem
Trace replay	signed frames, ordering, replay	no Internet latency benchmark
E2E tests	build, test, lint, doc, package	no symbolic crypto proof
CI witness		no benchmark regeneration

Table IV

Verification evidence and explicit limits. Entries are finite-scope implementation evidence; none is a wide-area Rings lookup benchmark.

Definition 38 (Latency decomposition).

$$\begin{aligned}
T_{total} &= T_{conn} + h \cdot T_{hop} + T_{crypto} + T_{app} + T_{witness}. \\
h &= O(\log |N|) \quad (\text{Inv}_{ring}(N, S) \wedge \text{FingerInv}(N, S)).
\end{aligned}$$

Symbol	Artifact anchor	Admission status
FiniteModel	dht_stateright.rs	admitted only for finite Γ
TraceReplay	trace_replay.rs	deterministic schedules
Tests _{CI}	qaci.yml	build/test witness, no benchmark regen
FrameImpl	message/e2e.rs	signed stream-frame behavior
ChunkImpl	chunk.rs	bounded whole-message reassembly

Table V

Artifact manifest for the current paper claim ledger. Each row names the repository anchor that can support an admitted finite-scope claim.

Symbol	Coordinate source	Ledger status
D_C^{paper}	Chord paper table	ExtBase _C , external baseline
D_C^{656}	PR 656 Chord-only dummy benchmark	ReportedOnly $\cap$ Baseline _C
D_R^{docker}	reported Docker/WebRTC run	ReportedOnly $\cap$ Runtime _R

Table VI

Reported coordinate ledger. These rows calibrate comparison or runtime context; they are not admitted artifact evidence until provenance is complete.

Only hT_{hop} is Chord-comparable; the rest is Rings runtime overhead.

Definition 39 (Benchmark data model).

$$\begin{aligned}
\eta &= (N, r, m, L, f, \lambda, \gamma, \mu_B), \\
\mu_B &\in \{\text{maintained}, \text{stale}\}, \\
\pi(d) &\in \text{Prov}, \\
\text{Regen}(d) &\Rightarrow \text{src}(d) \neq \perp \wedge \text{rev}(d) \neq \perp \\
&\quad \wedge \text{cmd}(d) \neq \perp \wedge \text{seed}(d) \neq \perp \\
&\quad \wedge \text{host}(d) \neq \perp.
\end{aligned}$$

$$\begin{aligned}
F_C(\eta; D) &\subseteq Q_C(\eta; D), \\
\phi_C(\eta; D) &= |F_C(\eta; D)| / |Q_C(\eta; D)|, \\
ok_C(\eta; D) &= 1 - \phi_C(\eta; D), \\
\bar{t}_C(\eta; D) &= |Q_C|^{-1} \sum_{q \in Q_C} t_C(q), \\
B_C(\eta; D) &= (\mathbb{E}[h_C^f \mid \eta, D], \mathbb{E}[h_C^r \mid \eta, D], \bar{t}_C, ok_C, \phi_C), \\
h_C^r &= h_C^f + 1.
\end{aligned}$$

$$\begin{aligned}
D_{\text{external}} &= D_C^{paper}, \\
D_{\text{reported}} &= D_C^{656} \uplus D_R^{docker}, \\
D_{\text{all}} &= D_{\text{external}} \uplus D_{\text{reported}}, \\
D_{\text{admit}} &= \{d \in D_{\text{all}} : \text{AdmittedEval}(d)\}, \\
\mathcal{F}_{II} &= \{0, .10, .20, .30, .40, .50\}, \\
\Lambda_{III} &= \{.05, .10, .15, .20, .25, .30, .35, .40\}, \\
D_C^{paper} &= \{C_{II}(f) : f \in \mathcal{F}_{II}\} \\
&\quad \uplus \{C_{III}(\lambda) : \lambda \in \Lambda_{III}\}, \\
D_{\text{maint}}^{656} &= \{\text{stable}, f_{.10}, f_{.20}, \gamma_{.05}, \gamma_{.40}\}, \\
D_{\text{stale}}^{656} &= \{f_{.10}, f_{.20}, \gamma_{.05}, \gamma_{.40}\}, \\
D_C^{656} &= (D_{\text{maint}}^{656} \times \{\text{maintained}\}) \\
&\quad \uplus (D_{\text{stale}}^{656} \times \{\text{stale}\}), \\
D_R^{docker} &= \{D_R^{\text{conv}}, D_R^{\text{tx}}\}.
\end{aligned}$$

$$\begin{aligned}
D_C^{paper} &: (N, r) = (1000, 20), \\
D_C^{656} &: (N, r, m, L) = (1600, 3, 160, 64), \\
D_C^{paper} &\subset \text{ExtBase}_C, \\
D_C^{656} &\subset \text{ReportedOnly} \cap \text{Baseline}_C, \\
D_R^{docker} &\subset \text{ReportedOnly} \cap \text{Runtime}_R, \\
\text{Baseline}_C \cap \text{Runtime}_R &= \emptyset.
\end{aligned}$$

Here D_C^{paper} is an external Chord-paper coordinate, D_C^{656} is a PR 656 Chord-only coordinate at Rings' current overlay parameters, and D_R^{docker} is reported Rings runtime evidence. The sets are disjoint by claim role: external baseline, reported Chord baseline, reported Rings runtime. Unless $\text{AdmittedEval}(d)$ holds, a tuple is a reported coordinate only; it is not used to prove $\text{Core}_{\text{Rings}}$ or any Rings performance claim. Metrics are not pooled unless a Rings lookup benchmark with the same workload key η is added.

Table VII records the Chord-paper baseline coordinates ($N = 1000, r = 20$). It calibrates comparison scale only; it is neither Rings runtime evidence nor generated by this repository.

η	N, r	$\mathbb{E}[h_C^f]$	timeout _C	10^4 fail_C
$C_{II}, f = 0$	1000, 20	3.815	0.000	0.0
$C_{II}, f = .10$	1000, 20	4.007	0.522	0.0
$C_{II}, f = .20$	1000, 20	4.177	1.071	0.0
$C_{II}, f = .30$	1000, 20	4.354	1.830	0.0
$C_{II}, f = .40$	1000, 20	4.791	3.700	0.0
$C_{II}, f = .50$	1000, 20	5.345	6.565	0.0
$C_{III}, \lambda = .05$	$\simeq 1000, 20$	3.905	0.076	7.0
$C_{III}, \lambda = .10$	$\simeq 1000, 20$	3.949	0.168	19.0
$C_{III}, \lambda = .15$	$\simeq 1000, 20$	4.044	0.272	18.0
$C_{III}, \lambda = .20$	$\simeq 1000, 20$	4.094	0.335	35.0
$C_{III}, \lambda = .25$	$\simeq 1000, 20$	4.128	0.405	36.0
$C_{III}, \lambda = .30$	$\simeq 1000, 20$	4.221	0.501	47.0
$C_{III}, \lambda = .35$	$\simeq 1000, 20$	4.308	0.599	60.0
$C_{III}, \lambda = .40$	$\simeq 1000, 20$	4.345	0.656	58.0

Table VII

External Chord-paper baseline coordinates ($N = 1000, r = 20$), transcribed as reported comparison coordinates. They are not Rings runtime evidence and are not regenerated by the current artifact.

Table VIII fixes a Chord-only baseline at Rings' current overlay parameters ($N = 1600, r = 3, m = 160, L = 64$). Maintained rows are the comparison baseline; stale rows intentionally disable maintenance and measure sensitivity, not Rings behavior.

η	model	$\bar{h}^f / \bar{h}^r$	ok	$\bar{t}_C$	$10^4 \phi$
stable	maint.	4.469/5.469	1.000	0.000	0.0
$f = .10$	maint.	4.394/5.394	1.000	0.000	0.0
$f = .20$	maint.	4.316/5.316	1.000	0.000	0.0
$\gamma = .05$	maint.	4.465/5.465	1.000	0.000	0.0
$\gamma = .40$	maint.	4.470/5.470	1.000	0.000	0.0
$f = .10$	stale	4.758/5.758	.990	.734	98.2
$f = .20$	stale	5.186/6.186	.969	1.890	314.7
$\gamma = .05$	stale	4.617/5.617	.945	.361	546.2
$\gamma = .40$	stale	5.746/6.746	.484	4.604	5161.1

Table VIII

Chord-only baselines at Rings parameters ($N = 1600, r = 3, m = 160, L = 64$). Maintained rows are the baseline; stale rows are no-maintenance sensitivity. These are PR 656 reported coordinates, not admitted current-artifact evidence until generator, revision, seed, command, and host are shipped.

Thus stale rows explain sensitivity, not the maintained Chord baseline:

$$\begin{aligned}
\mu_B = \text{stale} &\Rightarrow \neg \text{maintenance}, \\
\mu_B = \text{maintained} &\Rightarrow \text{Baseline}_C^{656}.
\end{aligned}$$

Obligation 6 (Benchmark report).

$$\begin{aligned}
\text{Report}_{\text{eval}} &= (\text{Baseline}_C^{paper}, \text{Baseline}_C^{656}, \\
&\quad \text{Runtime}_R^{docker}, \text{Prov}, \text{Limit}).
\end{aligned}$$

$$\begin{aligned}
\text{ReportCmp}_{lookup} &= B_C(\eta; D_C^{paper} \uplus D_C^{656}), \\
\text{Measure}_{runtime} &= B_R^{docker}(\eta_R; D_R^{docker}), \\
\text{ReportCmp}_{lookup} &\not\equiv \text{Measure}_{runtime}.
\end{aligned}$$

A Rings performance claim must measure Rings under the reported build; a Chord baseline may only be cited as a baseline.

$$\begin{aligned}
\text{AdmitPerf}_R(\eta) &\Leftrightarrow \exists D_R^{lookup}. \\
&\quad \text{AdmittedEval}(D_R^{lookup}) \\
&\quad \wedge \text{workload}(D_R^{lookup}) = \eta \\
&\quad \wedge \text{metric}(D_R^{lookup}) \supseteq \{h, T_{hop}, T_{conn}, a, \text{IPFS/libp2p}, \text{Kademlia}, \text{IPFS/libp2p}, \text{Pastry}, \text{Tapestry}, \text{CAN}\}, \\
&\quad T_{crypto}, T_{app}, T_{witness}\}, \\
\neg \text{AdmitPerf}_R(\eta) &\Rightarrow \text{PerfClaim}_R(\eta) \\
&\quad \in \text{Obligation}. \\
\vec{T}_{total} &= (T_{conn}, hT_{hop}, T_{crypto}, \\
&\quad T_{app}, T_{witness}), \\
\text{Ccmp}(\vec{T}_{total}) &= hT_{hop}, \\
\text{Rovh} &= (T_{conn}, T_{crypto}, T_{app}, \\
&\quad T_{witness}, \text{runtime}).
\end{aligned}$$

Only hT_{hop} is Chord-comparable. Connection establishment, cryptography, application handling, witness generation, and runtime scheduling are Rings overheads and require Rings measurements.

$$\begin{aligned}
D_R^{conv} &= \text{conv}(n = 16, T_{wall} = 309.532 \text{ s}, \\
&\quad \text{fail} = 0/16, \text{partial} = 3/16, \\
&\quad t_{p50} = .194, t_{p95} = 1.026, \text{msg}_s = 8058.6), \\
D_R^{tx} &= \{\text{tx}(1, 10.46), \text{tx}(16, 107.35), \\
&\quad \text{tx}(64, 90.36)\}.
\end{aligned}$$

$$\text{tx} : \text{KiB} \rightarrow \text{Mbps}, \quad \text{msg}_s : \text{messages/s}.$$

The Docker/WebRTC tuple is convergence evidence, not a steady-state lookup benchmark. D_R^{tx} is payload throughput; it must not be compared to Chord forwarding hops or timeout rates.

Artifact provenance.

$$\begin{aligned}
\text{ProvFull}(d) &\Leftrightarrow \text{src}(d) \neq \perp \wedge \text{rev}(d) \neq \perp \\
&\quad \wedge \text{cmd}(d) \neq \perp \wedge \text{seed}(d) \neq \perp \\
&\quad \wedge \text{host}(d) \neq \perp, \\
\text{AdmittedEval}(d) &\Leftrightarrow \text{Regen}(d) \wedge \text{ProvFull}(d).
\end{aligned}$$

No CI command currently regenerates Table VII, Table VIII, D_R^{conv} , or D_R^{tx} .

Limitations. The current artifact supports finite model/test evidence and reported baseline coordinates. It does not yet support claims about wide-area latency, NAT population behavior, adversarial churn beyond the scoped models, symbolic cryptographic security, or steady-state Rings lookup performance.

XI. Related Work

Self-certifying systems bind names to keys, while DID standardization separates identifier syntax, verification methods, and resolution metadata [11], [12]. Rings follows that line by making DID control a cryptographic predicate, but it places resources through $\Omega_N(X, H) =$

$(X, \text{owner}_N \circ H)$ rather than through a global certificate authority or ledger.

Protection systems and authorization logics make authority an explicit predicate; monads and algebraic effects make runtime effects typed [13]–[16]. Rings uses these tools as composition discipline: authority, placement, and effect emission live in separate carriers and meet only through typed morphism laws.

Structured overlays give different placement laws. Chord gives clockwise successor ownership; Kademlia/IPFS/libp2p give XOR or provider-record lookup; Pastry, Tapestry, and CAN explore prefix, proximity, and coordinate-space routing [4], [17]–[22]. Rings' novelty is not another lookup metric; it is the separation between the circular owner law, DID/session authority, CRDT-style storage merge, and browser/native transport interpretation.

Security work on DHTs shows why this separation must be explicit. Secure routing, Sybil, and Eclipse attacks require admission, identifier assignment, neighbor-set constraints, auditing, or redundant routing beyond ordinary DHT maintenance [23]–[26]. Rings therefore states Sybil, Eclipse, anonymity, and DoS resistance as policy obligations, not as consequences of CorrectChord.

Anonymity systems make traffic analysis a primary security objective. Tor uses onion routing; Nym adds mix-network style privacy infrastructure [27], [28]. Rings does not prove sender, receiver, or route unlinkability. Its relay layer is only a typed transport policy:

$$P_{relay} \not\equiv \text{Anon}.$$

Rings storage uses the join-semilattice CRDT case when merge is available and an explicit finalizer otherwise [6]. WebRTC and WebAssembly are runtime carriers, not overlay algorithms; WebRTC measurement work also suggests that connection setup, NAT traversal, and payload throughput must be reported separately from DHT hop count [29]–[31].

XII. Conclusion

Rings should be read as a bounded theorem plus a claim ledger for an implemented algebraic object, not as an informal bundle of peer-to-peer mechanisms:

$$\text{Rings} = (R, \mathcal{R}_N, \text{Set}/R, \Omega_N, T_A^E, \Pi, \mathbb{G}, \mathcal{X}).$$

$$\begin{aligned}
\text{Established} &= \text{Core}_{Rings} \wedge \text{LedgerDisc}_{Rings}, \\
\text{ScopedEv} &= \{a : a \in \text{Artifact}_R \\
&\quad \wedge \text{AdmittedArtifact}(a)\}, \\
\text{RepCoord} &= D_C^{paper} \uplus D_C^{656} \uplus D_R^{docker}, \\
\text{Obligation} &= \text{wideAreaBenchmark} \cup \text{adversarialLiveness} \\
&\quad \cup \text{fullReplicaRealization}, \\
\text{NonClaim} &= \text{NonClaim}_{Rings}.
\end{aligned}$$

The scientific contribution is the composition law and its claim ledger: CorrectChord supplies the overlay safety root, Ω_N lifts resource namespaces into live owners, protocol morphisms preserve placement, and runtime effects remain typed. Implementation evidence supports only the

entries admitted by Table V; coordinates in Table VI are not proof evidence. Benchmark and security claims outside that ledger remain obligations.

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